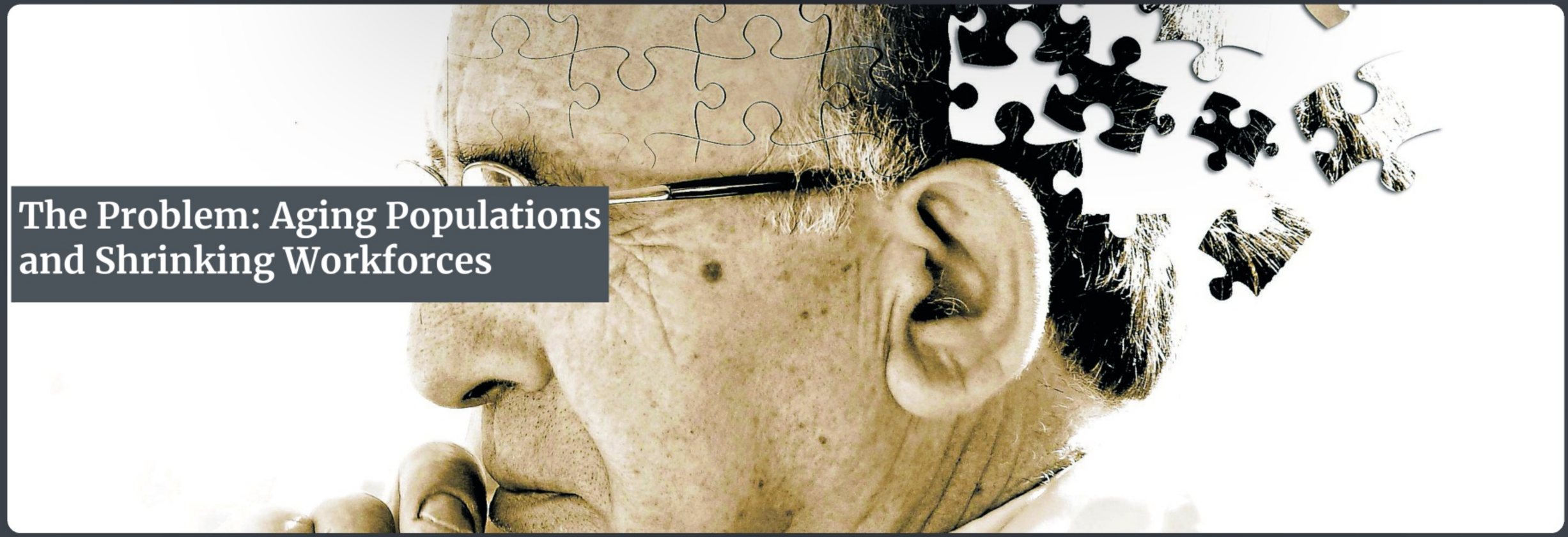
EncoreWorks: Supporting Longer Careers in Aging Societies

by Beatriz Fernandes, Alex Mateu and Rachel Vianna

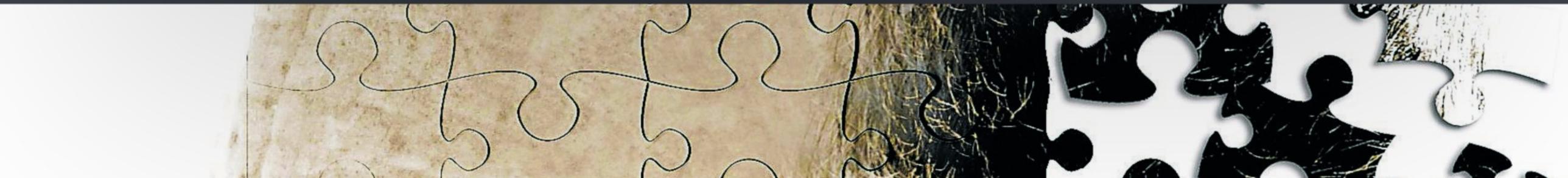


- Populations are aging rapidly across developed economies
- Working-age populations are shrinking
- Aging could reduce GDP growth by up to 40% in advanced economies
- Yet experienced workers often face age-related barriers in hiring, training, and career progression
- Unlocking the potential of workers aged 50+ could add up to $3.5 trillion to these economies



The Problem: Aging Populations and Shrinking Workforces

- Populations are aging rapidly across developed economies
- Working-age populations are shrinking
- Aging could reduce GDP growth by up to 40% in advanced economies
- Yet experienced workers often face age-related barriers in hiring, training, and career progression
- Unlocking the potential of workers aged 50+ could add up to $3.5 trillion to these economies



Product Overview: EncoreWorks

By exploring the relationship between workforce participation and economic outcomes, we identify strategies that support longer, more flexible careers.

These insights inform EncoreWorks, a SaaS platform that helps organizations manage and extend the careers of employees aged 50+, through workforce analytics, reskilling, and knowledge transfer.





Business Model

01. Key Features

Workforce Analytics Dashboard: analyzes age distribution, predicts retirement risk, and identifies critical skill gaps to foster career longevity.

Reskilling Programs: personalized training recommendations to help senior employees transition into new roles.

Knowledge Transfer Tools: mentoring programs and systems to capture and share expertise before retirement.

02. Customers

Large companies and HR departments in aging economies such as Japan, Germany, and Spain.



03. Revenue Model

Subscription-based SaaS model for companies, priced per employee or per organization, with additional revenue from premium analytics and training partnerships.

04. Value Proposition

Helps organizations reduce labor shortages, retain experienced talent, and maintain productivity in aging societies.

Hypotheses: Informing Our Business



Countries with aging populations can maintain economic productivity if older adults remain active in the workforce.

Countries that invest more in human capital have higher labor participation among older adults.

Data & Methodology

World Development Indicators Dataset (WDI)

The World Development Indicators dataset gives us detailed global data on the economy, population, and job markets for 200+ countries. This allowed us to gage aging populations and workforce trends.



Human Capital Index Dataset (HCI)

The Human Capital Index shows how well countries develop their people through education, skills, and health. This helped us understand the potential of populations, especially as they age.



Labor Force Participation Rate Indicator

The Labor Force Participation Rate measures the percentage of people of working age who are either employed or seeking a job. This gave us an outlook of how many people are economically active in the countries we looked at.



Limitations

- The Human Capital Index dataset, which is published periodically (every 3–5 years), currently provides comprehensive data only up to 2020.
- Because our analysis combines this dataset with World Development Indicators, we limited the study to the overlapping years of 2010 and 2018.
- Using an eight-year interval helps reduce short-term economic fluctuations and better capture structural relationships between human capital investment, demographic change, and economic outcomes. Since these indicators evolve gradually, this timeframe still provides a meaningful basis for comparing how countries manage aging workforces.



Key Findings: Insights from Data Analysis

Key Findings: Insights from Data Analysis

Aging is accelerating fastest in advanced economies such as Japan, Finland, and Czechia.

Key Findings: Insights from Data Analysis

Aging is accelerating fastest in advanced economies such as Japan, Finland, and Czechia.

Countries with older populations are not necessarily less productive — several maintain strong economic performance despite aging.

Key Findings: Insights from Data Analysis

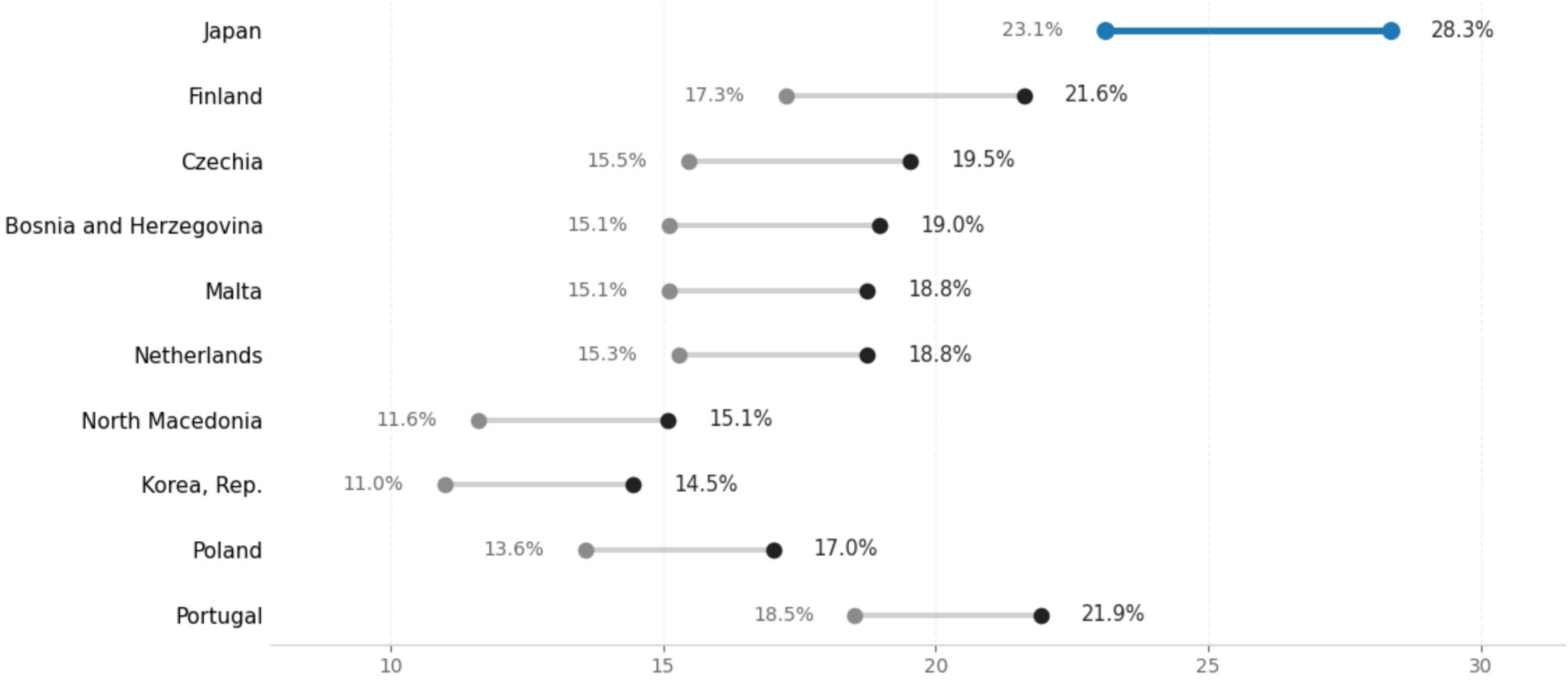
Aging is accelerating fastest in advanced economies such as Japan, Finland, and Czechia.

Countries with older populations are not necessarily less productive — several maintain strong economic performance despite aging.

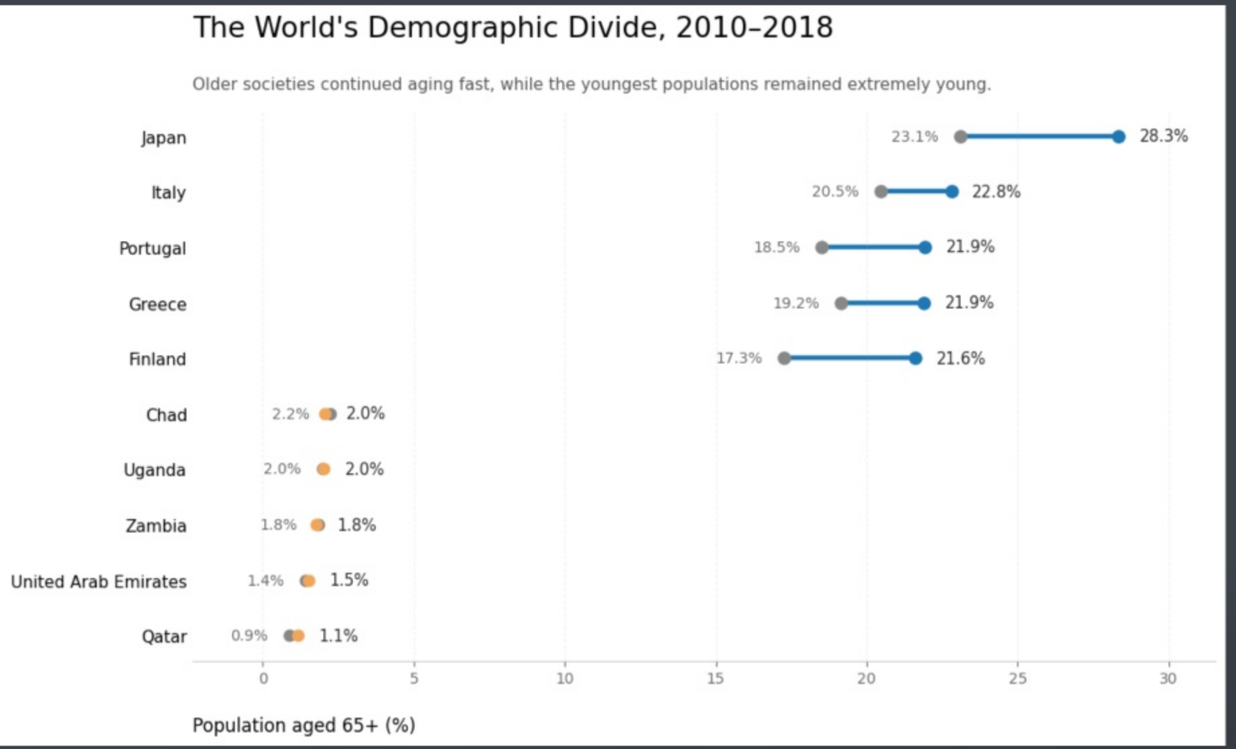
Human capital investment appears strongly linked to workforce participation, suggesting skills development plays a key role in sustaining productivity.

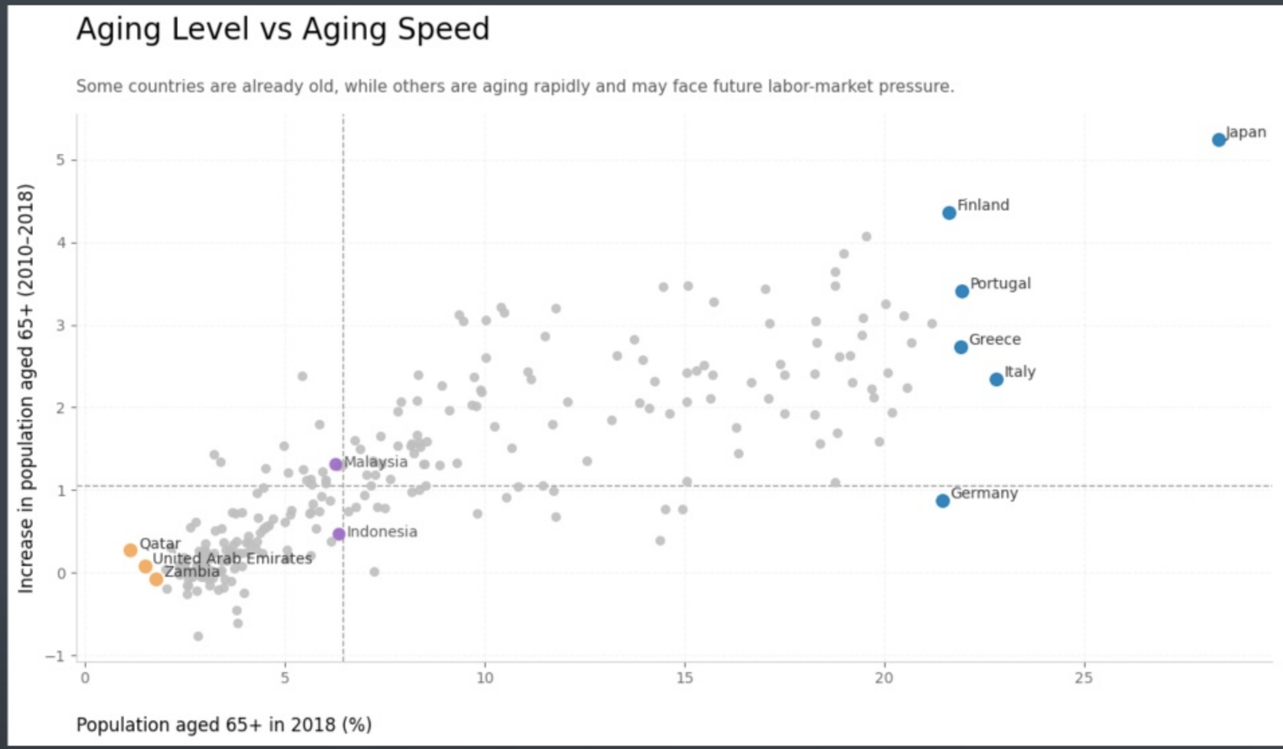
Top 10 Fastest Aging Societies, 2010–2018

Share of population aged 65+ increased fastest in Japan, Finland, and Czechia in the span of 8 years.



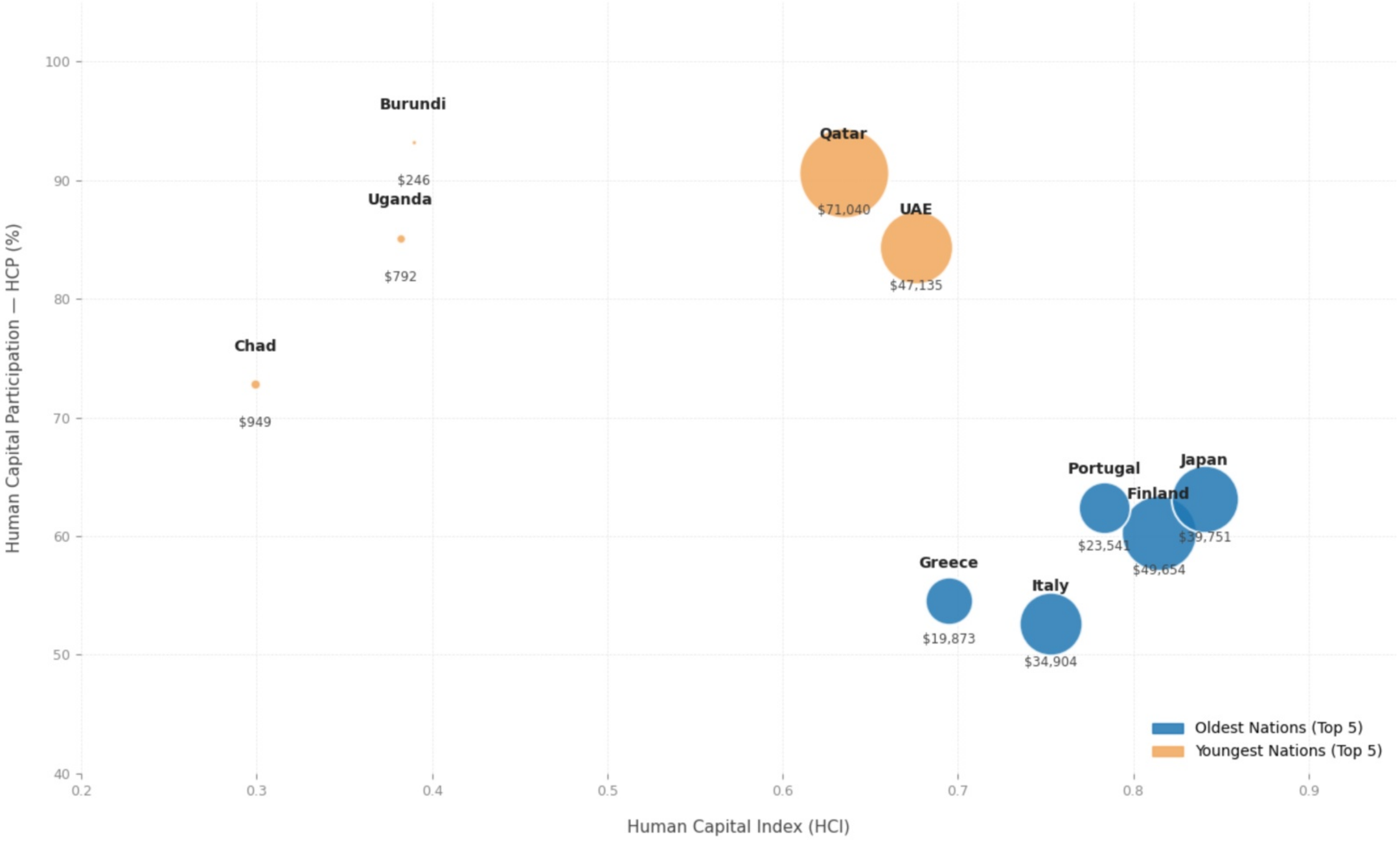
Population aged 65+ (%)


The World's Demographic Divide, 2010–2018
Older societies continued aging fast, while the youngest populations remained extremely young.
Japan
Italy
Portugal
Greece
Finland
Chad
Uganda
Zambia
United Arab Emirates
Qatar
23.1%
28.3%
20.5%
22.8%
18.5%
21.9%
19.2%
21.9%
17.3%
21.6%
2.2%
2.0%
2.0%
2.0%
1.8%
1.8%
1.4%
1.5%
0.9%
1.1%
0
5
10
15
20
25
30
Population aged 65+ (%)


Aging Level vs Aging Speed
Some countries are already old, while others are aging rapidly and may face future labor-market pressure.
Increase in population aged 65+ (2010–2018)
Japan
Finland
Portugal
Greece
Italy
Malaysia
Germany
Indonesia
Qatar
United Arab Emirates
Zambia
5
4
3
2
1
0
−1
0
5
10
15
20
25
Population aged 65+ in 2018 (%)

GDP, Human Capital Index vs Workforce Participation

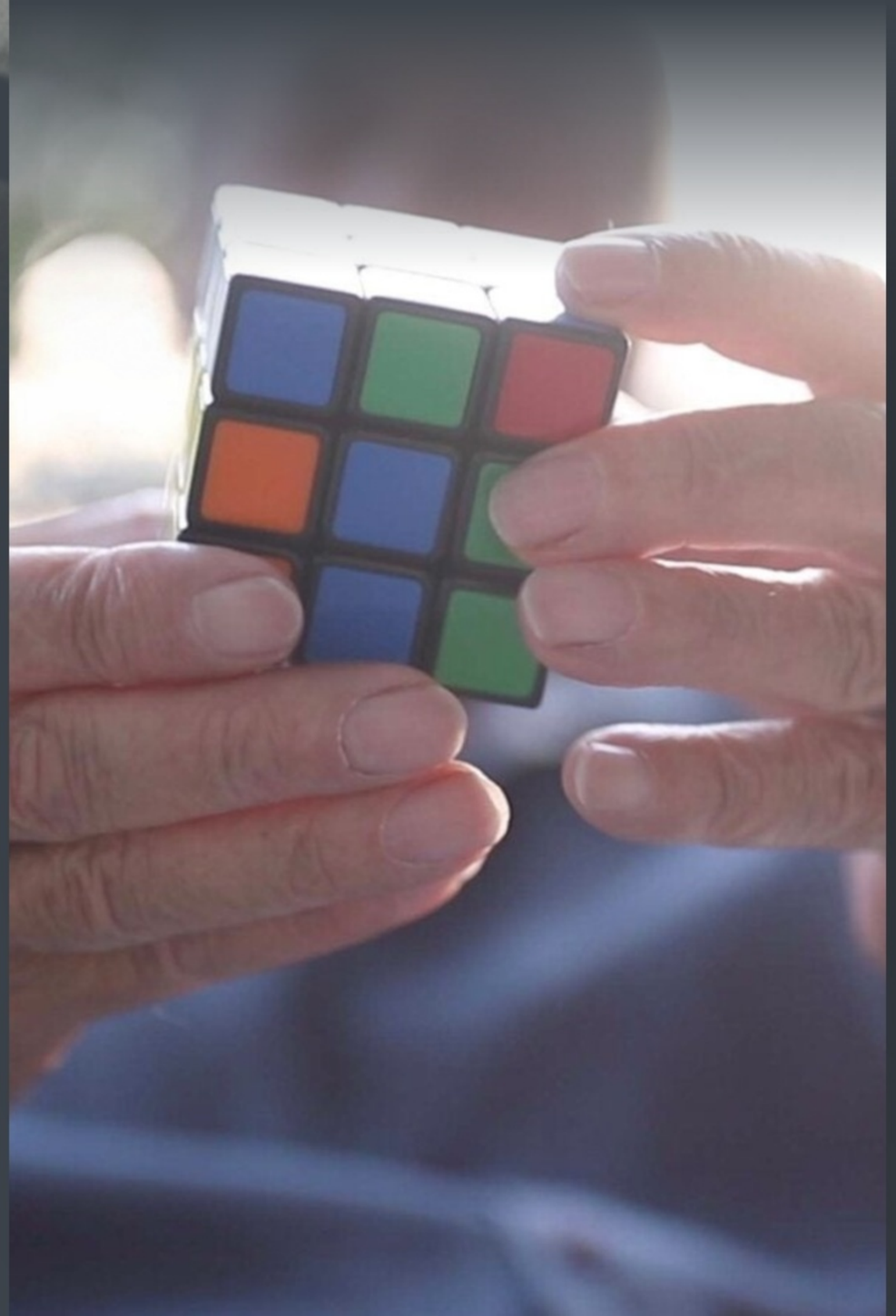
Bubble size represents GDP per capita (2018). Older nations show higher skills, younger nations higher participation.



Conclusion: The Real Aging Challenge It's Keeping Workers Skilled

- High human capital investment and high labor participation reflect two different development paths to economic output.
- Productivity-led (oldest nations): fewer workers, higher skills, higher GDP per worker
- Participation-led (youngest nations): more workers, lower skills, lower GDP per worker





Recommendations: Embracing Productive Aging

- The risk for aging nations isn't workforce shrinkage, it's skill erosion and early retirement
- The risk for young nations isn't too many workers, it's low productivity per worker
- Policy implication: investing in lifelong learning and human capital quality is more impactful than raising retirement age alone



Sustaining economic productivity in aging societies require quality of investment in human capital.
Not higher quantity of workers.

GDP per capita requires a minimum threshold of human capital quality — no amount of labor participation compensates for low HCI.

Thank You

from Beatriz Fernandes, Alex Mateu
and Rachel Vianna

Investing in human capital is no longer optional –
it is a macroeconomic necessity.





Take this with you.
Revisit anytime.

Missed something? Want to explore further?
Scan or click below to open this presentation.
Anytime, anywhere.

View presentation

